

Design and Analysis of Algorithms

Course Code	CSE 204	Course Category	CC		L	T	P	C
					3	0	1	4
Pre-Requisite Course(s)	CSE 102	Co-Requisite Course(s)		Progressive Course(s)				
Course Offering Department	CSE	Professional / Licensing Standards						

Course Objectives / Course Learning Rationales (CLR s)

1. To impart basic skills to analyse the performance of algorithms.
2. To train the students to choose appropriate algorithm design techniques for solving problems.
3. To make aware how the choice of data structures and algorithm design methods impact the performance of programs.
4. To impart basic proficiency to deal with NP problems and to develop approximate algorithms wherever required
5. To create an understanding of the basic issues of complex and efficient algorithms.
6. To introduce advanced topics of Backtracking and Branch and bound algorithms required in state space search.

Course Outcomes / Course Learning Outcomes (CLOs)

	At the end of the course, the learner will be able to	Bloom's Level	Expected Proficiency Percentage	Expected Attainment Percentage
Outcome 1	Choose appropriate algorithm design techniques for solving problems.	4	70%	65%
Outcome 2	Describe how the choice of data structures and algorithm design methods impact the performance of programs.	2	70%	65%
Outcome 3	Analyse the performance of algorithms.	4	70%	65%
Outcome 4	Develop approximate algorithms with NP problems.	4	70%	65%
Outcome 5	Explain the complexity and efficiency of algorithms.	3	70%	65%
Outcome 6	Demonstrate Backtracking, Branch and bound algorithms required in state space search.	4	70%	65%

Course Articulation Matrix (CLO) to Program Learning Outcomes (PLO)

CLOs	Program Learning Outcomes (PLO)														
	Engg in eeri ngn Kn owl ed ge	Pro blem Ana ly sis	Desi gn and Deve lop ment	Analysis, Design and Reso lution	Mod eling and Sim ulation	Soci ety and Mult icultural Skill s	En viron men tal and Sus tain able	Mo ral and Eth ical Awareness	Indi vidual and Team work Skill s	C ommu nity and Soc iety	Proje ct Man age ment and Lea dership	Self - Directed and Long term Learning	PSO 1	PSO 2	PSO 3

				h			n a b i l i t y	n e s s			n c e				
Outcome 1	2	3	3	3	3	1			2		2	2	2	2	2
Outcome 2	2	2	3	2	2	1			2		2	3	2	2	2
Outcome 3	2	3	3	3	2	1			2		2	2	2	2	2
Outcome 4	3	3	3	2	3	1			2		3	3	3	2	3
Outcome 5	3	3	3	3	2	1			2		2	3	2	2	2
Outcome 6	3	3	3	3	2	1			2		2	2	3	3	2
Average	3	3	3	3	2	1			2		2	3	2	2	2

Unit No.	Unit Name	Required learning hours	CLOs Addressed	References Used
UNIT I	Introduction	9		
	Algorithmic thinking & motivation with examples	1	1,3	1
	Reinforcing the concepts of Data Structures with examples	2	1,4	1,2
	Complexity analysis of algorithms: big O, omega, and theta notation	2	2	1
	Analysis of Sorting and Searching	2	2	2
	Hash table	1	4	1
	Recursive and non-recursive algorithms.	1	4	1
	General Problem Solving (GPS) techniques	9		
Unit II	Divide and conquer: Merge sort	2	1,3	1
	Quicksort		1,3	1,2
	BST	1	1,3	1,2
	Master method for Complexity analysis		2	1,2
	Greedy method: Fractional Knapsack	2	3,4	1
	Minimum spanning trees (Prim's & Kruskal's)		4	1,2
	Shortest paths: Dijkstra's algorithm	2	4	1,2
	Huffman coding		4	1,2
	Dynamic Programming: 0/1 Knapsack	2	1,4	1,2
	All-to-all shortest paths		4	1,2
	Search techniques and Randomised algorithms	9		
	BFS & DFS, Backtracking	1	2,4	1
UNIT III	8-Queen's problem	1	4	
	Knight's tour	1	4	1
	Travelling Salesman Problem (TSP)	1	3,4	1
	Branch-and-bound: 16-puzzle problem	1	4	1
	TSSP	2	4	1
	Randomized algorithms: Playing Cards	2	4	2,3
	Pattern matching and Amortized analysis	9		
	Pattern matching algorithms: Brute-force,	2	4	4
	Boyer Moore	2	4	4
	KMP algorithms	1	3,4	4
UNIT IV	Algorithm analysis: Probabilistic Analysis	2	2	4
	Amortized analysis,	1	2	4
	Competitive analysis	1	2	4
	NP problems	9		
	Non-polynomial complexity: examples and analysis	2	4	2,4
	Vertex cover	1	3,4	2
	Set cover	1	4	2,4
	TSP	1	4	2,4
	3-SAT	1	4	2,4
	Approximation Algorithms: Vertex cover	1	4	2,4
UNIT V	TSP	1	4	2,4

	Set cover	1	4	2,4
Total contact hours		45		

Course Unitization Plan La b

Exp No.	Experiment Name	Required learning hours	CLOs Addressed	References Used
1	Programs for summation of series $1+X+X^2+X^3+\dots$ with different time complexities	2	1	
2	Converting recursive programs to non-recursive programs. Towers of Hanoi Problem example.	2	1	
3	Implementation of Bucket Sort with time complexity analysis	2	2	
4	Divide and conquer: Implementation of maximum-subarray problem.	2	2	
5	Divide and conquer: Implementation of Strassen's algorithm for matrix multiplication	2	2	
6	Greedy Approach: Implementation of Fractional Knapsack	2	3	
7	Greedy Approach: Implementation of Task-scheduling problem.	2	3	
8	Greedy Approach: Implementation of Huffman Code.	2	3	
9	Dynamic Programming: Implement 0/1 Knapsack problem	2	3	
10	Implementation of (a). Find connected components and cycles in undirected graphs. (b). Topological sort	2	4	
11	Implementation of (c). Cycles in directed graphs. (d). Biconnected Components.	2	4	
12	Implementation of (e.) Bellman Ford (f). Ford-Fulkerson method.	2	4	
13	Branch and Bound: (a). Implementation of 8 Queens problem (b). Implementation of 15-puzzle problem	2	5	
14	Pattern matching: (a). Implementation of Boyer Moore algorithm (b). Implementation of KMP algorithm	2	5	
15	Implementation of Approximation algorithms: TSP	2	5	
	Total Hours		30	

Learning Assessment Theory

Bloom's Level of Cognitive Task		Continuous Learning Assessments (30%)				End Semester Assessments (30%)
		CLA-1 (6%)	Mid-1 (12%)	CLA-2 (6%)	CLA-3 (6%)	
Level 1	Remember	60%	30%	30%	30%	30%
	Understand					
Level 2	Apply	40%	70%	70%	70%	70%
	Analyse					
Level 3	Evaluate					
	Create					
Total		100%	100%	100%	100%	100%

Learning Assessment Theory

Bloom's Level of Cognitive Task		Lab Performance (20%)	End Semester Exam (20%)
Level 1	Remember	30%	30%
	Understand		
Level 2	Apply	70%	70%
	Analyse		
Level 3	Evaluate		
	Create		

Total	100%	100%
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Recommended Resources

1. Cormen, T. H., Leiserson, C. E., Rivest, R. L., & Stein, C. (2022). Introduction to algorithms. MIT press.
2. Dave, P. H., & Dave, H. B. (1900). Design and analysis of algorithms. Pearson Education India.
3. Goodrich, M. T., & Tamassia, R. (2001). Algorithm design: foundations, analysis, and internet examples. John Wiley & Sons.
4. Aho, A. Hopcroft, J. E. Ullman, J. D. Design and Analysis of Algorithms, Addison-Wesley Publishing.

Other Resources

1. Kleinberg, J. and Tardos, E. (2005) Algorithm Design Addison-Wesley.**Other Resources**